ENES 246 · DIGITAL LOGIC DESIGN

Lecture 20

State Reduction and Sequential Design



Equivalent states · state reduction · state assignment · designing with D, JK and T flip-flops · unused states

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapter 5, §5.7–5.8

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Today’s roadmap

1 §5.7

State reduction

Merge equivalent states

2 §5.8

State assignment

Binary, Gray or one-hot

3 §5.8

Design with D

Table → maps → circuit

4 §5.8

JK and T design

Excitation tables

By the end of class you can…

- Find equivalent states by row matching and the implication table
- Choose a state assignment and count flip-flops
- Design a sequence detector with D flip-flops
- Use excitation tables to design with JK and T flip-flops
- Check that unused states cannot trap the circuit

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Agenda for today's session



Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome, roadmap	—
0:05 – 0:30	Equivalent states; implication table	Example 1
0:30 – 0:40	State assignment	—
0:40 – 0:50	Break	
0:50 – 1:15	The design procedure with D flip-flops	Example 2
1:15 – 1:40	Excitation tables; JK and T design	Examples 3, 4 · Your turn
1:40 – 1:50	Unused states; concept check; wrap-up	—

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Equivalent states

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Definition

Two states are equivalent when, for every input, they give the same output and go to equivalent next states.

Why reduce?

Fewer states can mean fewer flip-flops ($2^n \geq$ states) and simpler logic, with identical input-output behavior.

Two methods

Row matching: merge identical rows, repeat.

Implication table: finds every equivalence.

Row matching

1. Look for two rows with the same next states and outputs.
2. Keep one; replace the other everywhere in the table.
3. Repeat: merging can make new rows identical.

Stops when no rows match; the implication table may still find more.

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Example 1 — Reduce a state table

EXAMPLE

Reduce this seven-state table. Columns: next state and output for $x = 0$ and $x = 1$.

State	Next x=0	Next x=1	y x=0	y x=1
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	g	f	0	1
g	a	f	0	1

State	Next x=0	Next x=1	y x=0	y x=1
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	d	0	1
e	a	d	0	1

Reduced: five states

Equivalent pairs: $e \equiv g$, then $d \equiv f$. Every step is on the next slide.

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Example 1 — Step by step

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1. Equivalent: same outputs, and equivalent next states for each x
2. Group by outputs: $\{a, b, c\}$ give 0, 0; $\{d, e, f, g\}$ give 0, 1
3. In the second group: $e \to (a, f)$ and $g \to (a, f)$: identical
4. So $e \equiv g$. Replace g by e everywhere
5. Now $f \to (e, f)$ and $d \to (e, f)$: identical
6. So $d \equiv f$. Replace f by d: $d \to (e, d)$, $e \to (a, d)$
7. Check the first group: $a \to (a, b)$, $b \to (c, d)$, $c \to (a, d)$ differ
8. Repeat until no rows match: nothing more merges
9. Result: $\{a\}, \{b\}, \{c\}, \{d, f\}, \{e, g\}$: **five states**
10. Still 3 flip-flops, but 3 unused codes become don't-cares

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The implication table

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	a	b	c	d	e	f
b	X					
c	X	X				
d	X	X	X			
e	X	X	X	X		
f	X	X	X	eg	X	
g	X	X	X	X	✓	X

Procedure

One cell per pair of states.

X if the outputs differ.

Otherwise list the implied pairs (next states that must also be equivalent); ✓ if none.

Pass again: X any cell whose implied pair is X. Repeat.

Unmarked cells are equivalent: here (d, f) and (e, g).

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State assignment

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State	Binary	Gray	One-hot
a	000	000	00001
b	001	001	00010
c	010	011	00100
d	011	010	01000
e	100	110	10000

Trade-offs

Binary: fewest flip-flops, $\lceil \log_2 m \rceil$.

Gray: states that follow each other differ in one bit; K-map groups tend to be larger.

One-hot: one flip-flop per state; next-state logic is trivial and fast (common in FPGAs).

Unused codes (3 here in binary) are don't-cares.

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Break

10 minutes — we resume at the 0:50 mark

Next: designing sequential circuits

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The design procedure

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1. From the word description, draw the state diagram.
2. Reduce the number of states if possible.
3. Assign binary codes: n flip-flops for up to 2^n states.
4. Choose the flip-flop type (D is the simplest to design with).
5. Build the encoded state table: present, input, next, output.
6. Derive and simplify each flip-flop input and output equation.
7. Draw the circuit; check any unused states.

Flip-flop inputs

D: $D = Q(t+1)$, read straight from the table.

JK: look up J and K in the excitation table.

T: $T = Q \oplus Q(t+1)$.

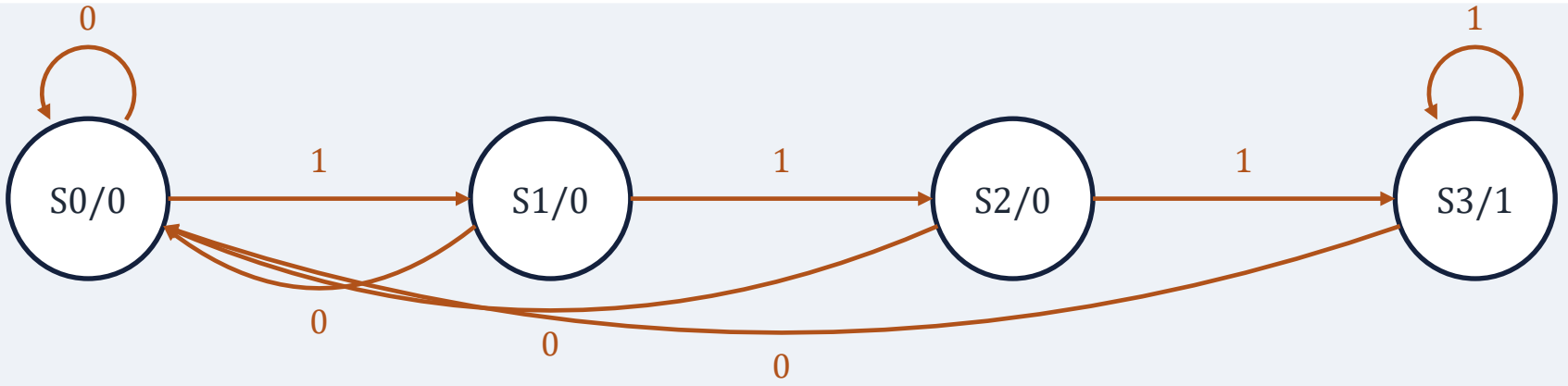
Unused states and excitation X entries are don't-cares.

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Example 2 — A sequence detector

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EXAMPLE Output $y = 1$ after three or more consecutive 1s on input x. Use D flip-flops.



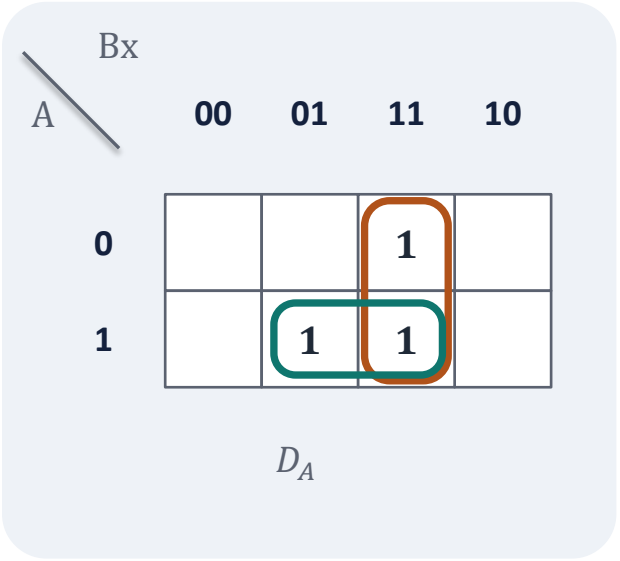
1. State S_k: the last k inputs were 1 ($k \leq 3$)
2. From any state, $x = 0$ → back to S_0
3. $x = 1$ → S_{k+1}; S_3 stays in S_3
4. Moore: $y = 1$ only in S_3
5. Four states → two flip-flops A, B
6. Codes: $S_0 = 00$, $S_1 = 01$, $S_2 = 10$, $S_3 = 11$

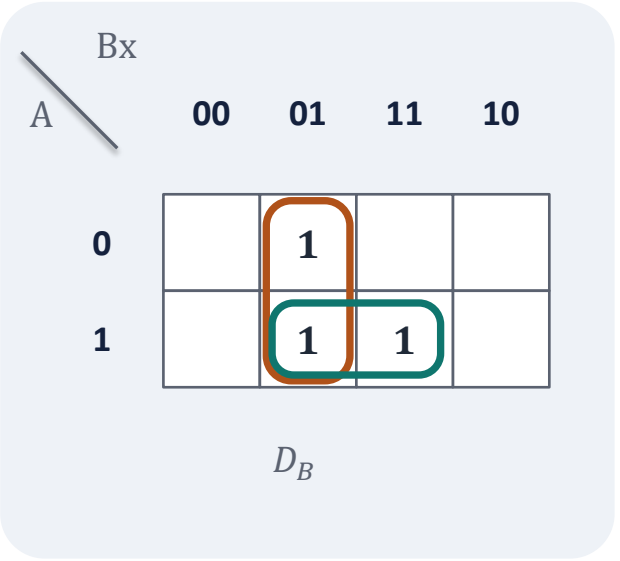
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Example 2 — Table and maps

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A B	x	A^+ B^+	y
0 0	0	0 0	0
0 0	1	0 1	0
0 1	0	0 0	0
0 1	1	1 0	0
1 0	0	0 0	0
1 0	1	1 1	0
1 1	0	0 0	1
1 1	1	1 1	1



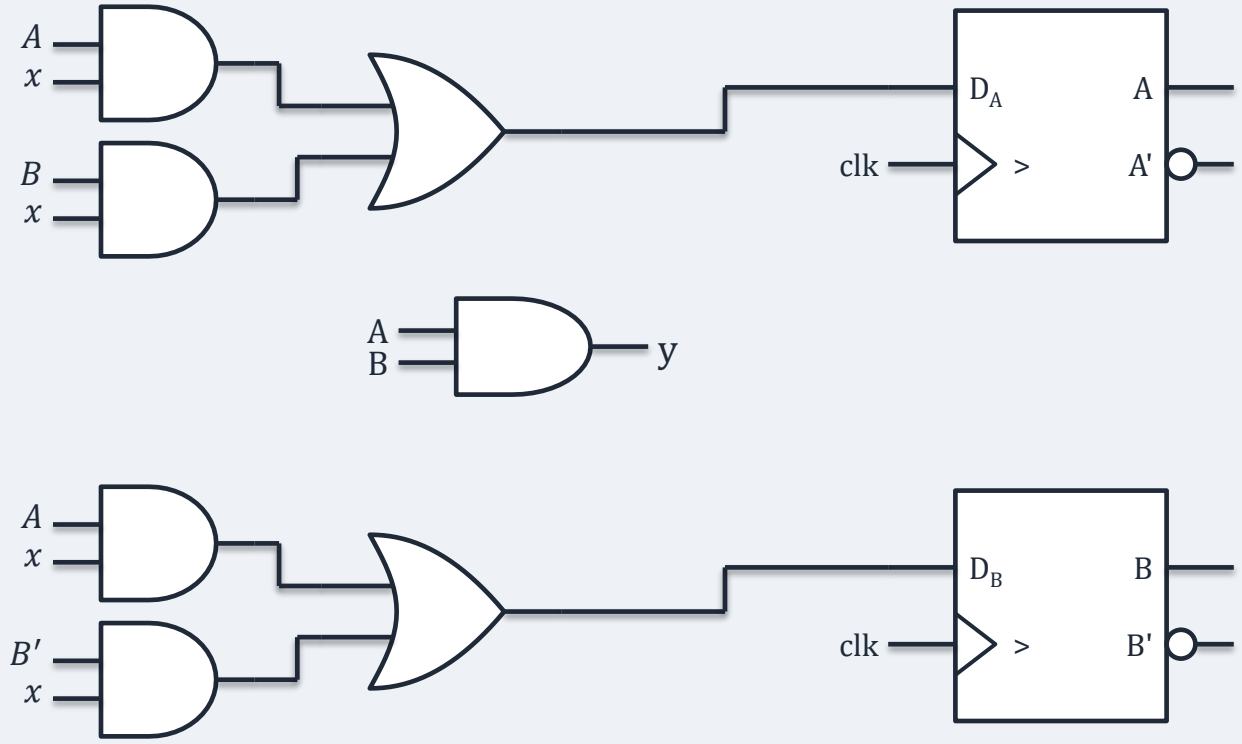


1. Next states from the diagram
2. D flip-flops: $D_A = A^+$, $D_B = B^+$
3. D_A: m3, m7 → Bx; m5, m7 → Ax
4. $D_A = Ax + Bx$
5. D_B: m1, m5 → $B'x$; m5, m7 → Ax: $D_B = Ax + B'x$
6. $y = 1$ only in 11: $y = AB$

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Example 2 — The circuit

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Check a sequence

x = 0 1 1 1 1 0

state: 00 → 00 → 01 → 10 → 11 → 11 → 00

y = 0 0 0 0 1 1 0

Feedback signals A, B, B' are drawn as labels to keep the figure readable.

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Excitation tables

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Q → Q⁺	J	K
0 → 0	0	X
0 → 1	1	X
1 → 0	X	1
1 → 1	X	0

Q → Q⁺	T
0 → 0	0
0 → 1	1
1 → 0	1
1 → 1	0

Q → Q⁺	D
0 → 0	0
0 → 1	1
1 → 0	0
1 → 1	1

Reading an excitation table

It answers: what input makes the flip-flop go from Q to Q^+? For JK, each transition has two ways to happen, so one input is a don’t-care: JK designs usually need the fewest gates.

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Example 3 — Design with JK flip-flops

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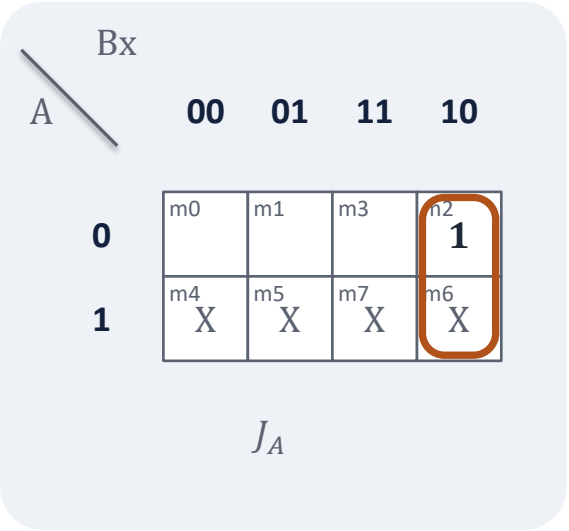
A B	x	A^+ B^+	JA	KA	JB	KB
0 0	0	0 0	0	X	0	X
0 0	1	0 1	0	X	1	X
0 1	0	1 0	1	X	X	1
0 1	1	0 1	0	X	X	0
1 0	0	1 0	X	0	0	X
1 0	1	1 1	X	0	1	X
1 1	0	1 1	X	0	X	0
1 1	1	0 0	X	1	X	1

1. Write each present state, input and next state
2. Excitation: $0 \to 0: J = 0, K = X;\ 0 \to 1: J = 1, K = X$
3. $1 \to 0: J = X, K = 1;\ 1 \to 1: J = X, K = 0$
4. Fill J_A, K_A from $A \to A^+$; J_B, K_B from $B \to B^+$
5. Map each column; every X is a don’t-care (next slide)
6. $J_A = Bx',\ K_A = Bx$
7. $J_B = x,\ K_B = A'x' + Ax$
8. $K_B = A'x' + Ax = (A \oplus x)'$, one XNOR

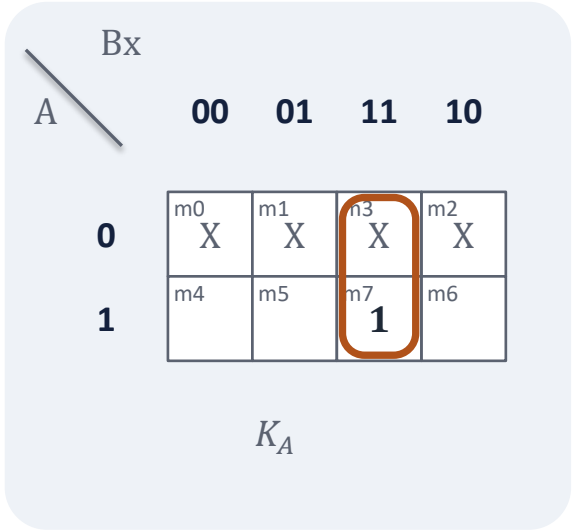
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Example 3 — The four K-maps

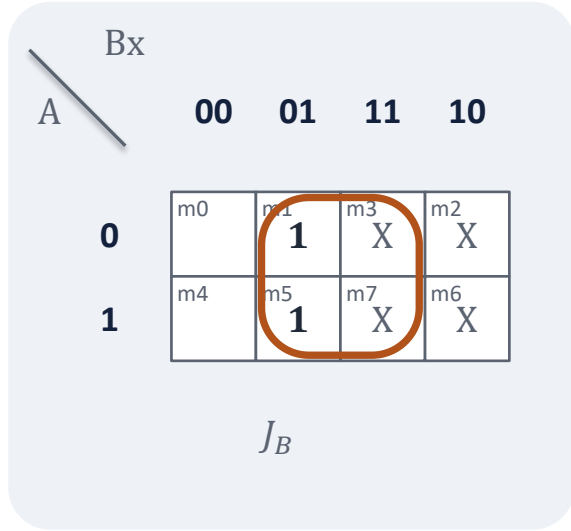
§5.8



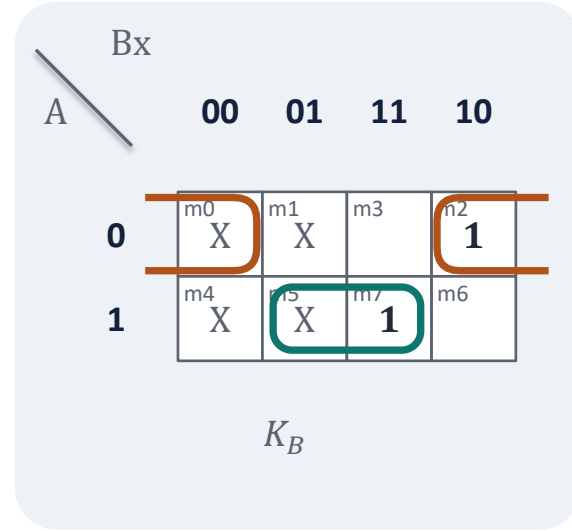
J_A



K_A



J_B



K_B

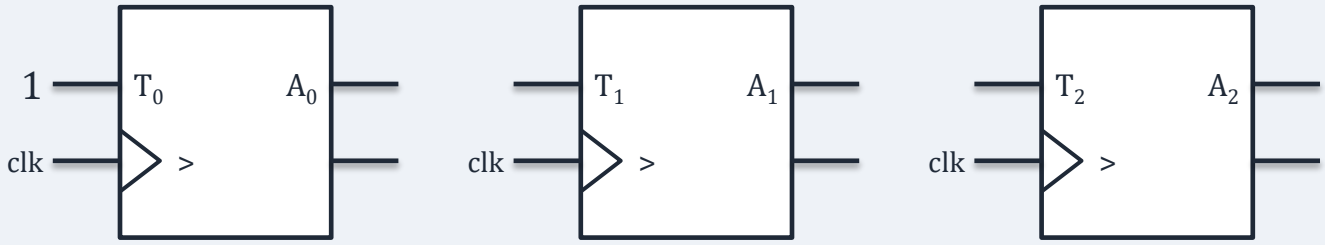
1. 1 at m2: A goes $0 \to 1$ (01, $x = 0 \to 10$)
2. X wherever $A = 1$: m4–m7
3. Pair m2 with the X at m6 → Bx'
4. $J_A = Bx'$
5. 1 at m7: A goes $1 \to 0$ (11, $x = 1 \to 00$)
6. X wherever $A = 0$: m0–m3
7. Pair m7 with the X at m3 → Bx
8. $K_A = Bx$
9. 1 at m1, m5: B goes $0 \to 1$
10. X wherever $B = 1$: m2, m3, m6, m7
11. Quad $x = 1$: m1, m3, m5, m7 → x
12. $J_B = x$
13. 1 at m2, m7: B goes $1 \to 0$
14. X wherever $B = 0$: m0, m1, m4, m5
15. m2 + X m0 → $A'x'$; m7 + X m5 → Ax
16. $K_B = A'x' + Ax = (A \oplus x)'$

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Example 4 — Counter with T flip-flops

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$A_2A_1A_0$	next	T_2	T_1	T_0
000	001	0	0	1
001	010	0	1	1
010	011	0	0	1
011	100	1	1	1
100	101	0	0	1
101	110	0	1	1
110	111	0	0	1
111	000	1	1	1



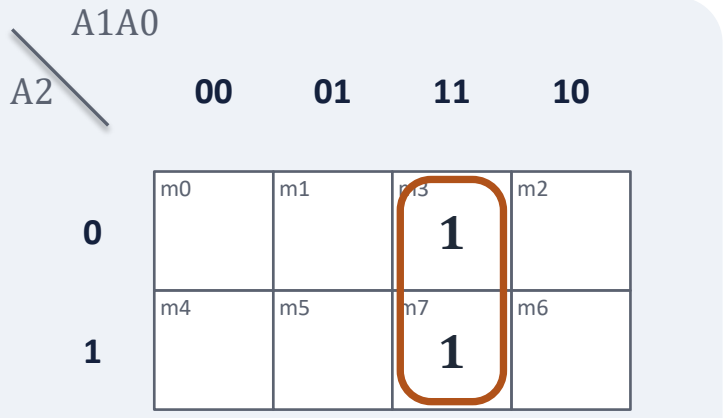
$$T_0 = 1,\ \ T_1 = A_0,\ \ T_2 = A_1A_0$$

1. $T = Q \oplus Q^+$: 1 where the bit changes
2. T_0 is 1 on every row → $T_0 = 1$
3. $T_1 = 1$ on 001, 011, 101, 111 → $T_1 = A_0$
4. $T_2 = 1$ on 011, 111 → $T_2 = A_1A_0$
5. A bit toggles when every lower bit is 1
6. Counts 000 → 001 → … → 111 → 000

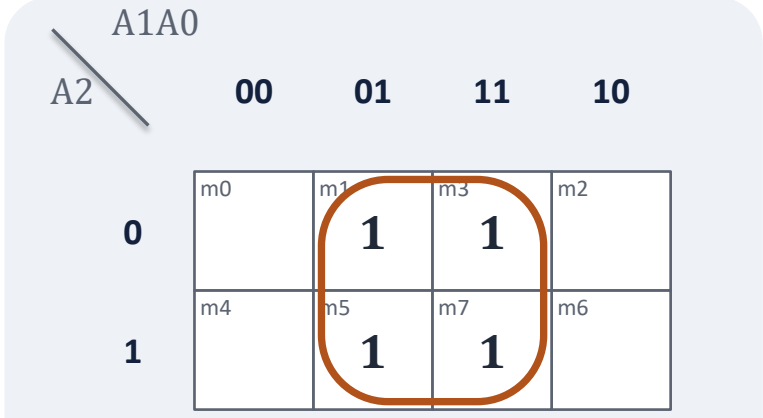
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Example 4 — The three K-maps

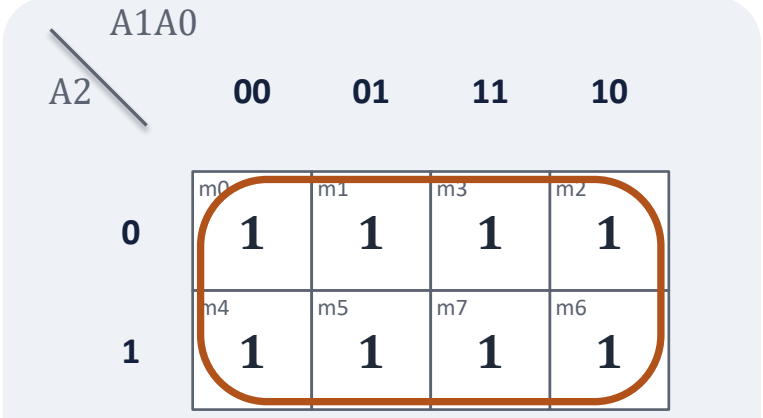
§5.8



$T_2 = A_1 A_0$



$T_1 = A_0$



$T_0 = 1$

1. 1 where A_2 changes: 011 → 100 and 111 → 000
2. m3 and m7 pair: $A_1 = A_0 = 1$
3. 1 where A_1 changes: m1, m3, m5, m7
4. The quad where $A_0 = 1$
5. A_0 changes on every count: all 8 cells are 1
6. The whole map is one group

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Your turn — Counter with enable

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YOUR TURN

≈ 12 min

A 2-bit counter AB counts 00 → 01 → 10 → 11 → 00 when $x = 1$ and holds when $x = 0$.

(a) Write the state table.

(b) Design it with D flip-flops.

(c) Design it with T flip-flops. Which needs fewer gates?

Map variables in the order A, B, x.

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Your turn — Solution

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SOLUTION

A B	x	A^+ B^+	T_A	T_B
0 0	0	0 0	0	0
0 0	1	0 1	0	1
0 1	0	0 1	0	0
0 1	1	1 0	1	1
1 0	0	1 0	0	0
1 0	1	1 1	0	1
1 1	0	1 1	0	0
1 1	1	0 0	1	1

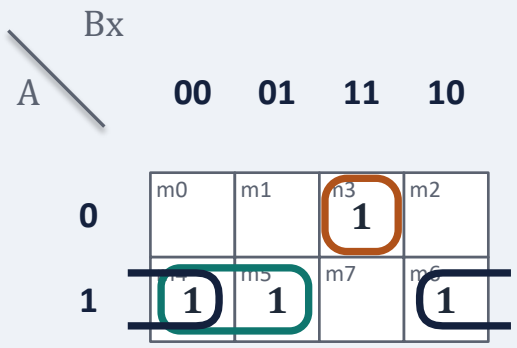
1. (a) $x = 0$: next = present; $x = 1$: next = present + 1
2. (b) D: $D = Q^+$. B changes only when $x = 1 \rightarrow D_B = B \oplus x$
3. A changes when $x = 1$ and $B = 1 \rightarrow D_A = A \oplus Bx$
4. Check 01, $x = 1$: $D_B = 1 \oplus 1 = 0$, $D_A = 0 \oplus 1 = 1 \rightarrow 10$ ✓
5. (c) T: $T = Q \oplus Q^+$. The T_B column equals $x \rightarrow T_B = x$
6. $T_A = 1$ only on 01/1 and 11/1 $\rightarrow T_A = Bx$
7. T needs one AND gate; D needs two XORs and an AND
8. $x = 0$: every $T = 0 \rightarrow$ hold ✓

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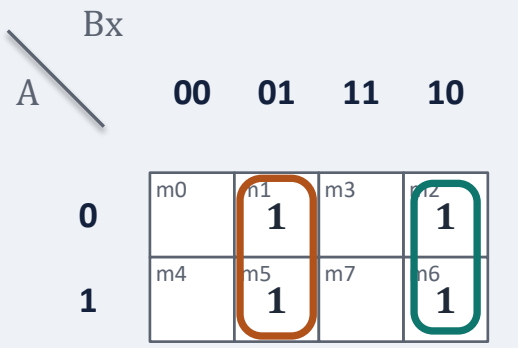
Your turn — The K-maps

§5.8

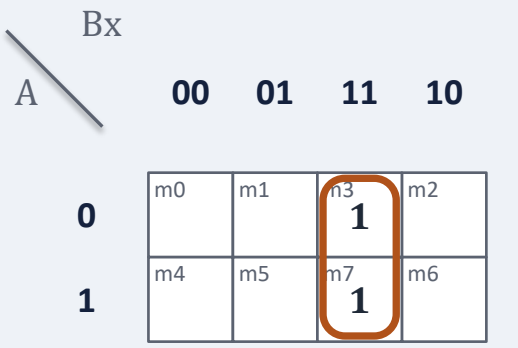
SOLUTION



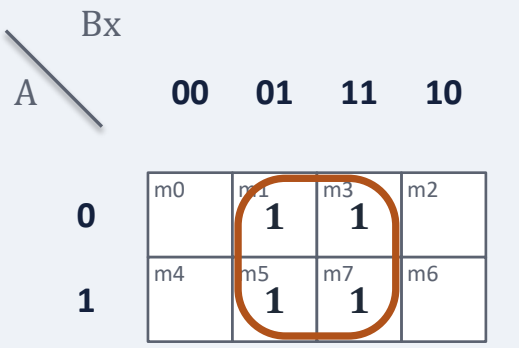
$D_A = A'Bx + AB' + Ax'$



$D_B = B'x + Bx' = B \oplus x$



$T_A = Bx$



$T_B = x$

1. D: D = next state. D_A needs 3 groups; D_B is a checkerboard
2. $D_A = A(B' + x') + A'Bx = A \oplus Bx$: two XORs and an AND in all
3. T: $T = Q \oplus Q^+$. T_A is one pair, T_B one quad
4. T needs one AND gate: the cheaper design

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Unused states

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The risk

Codes that are never assigned (e.g. 101, 110, 111 in a 5-state machine) were treated as don’t-cares.

At power-up or after noise, the circuit may land in one.

The check

Evaluate the final equations from every unused code for every input. Each path must reach a valid state.

If one loops among unused codes, change a don’t-care choice or add a reset.

Self-correcting

A circuit whose unused states all lead back into the valid sequence is self-correcting.

Counters in Lectures 21–22 are checked this way.

Five states need $\lceil \log_2 5 \rceil = 3$ flip-flops, leaving $8 - 5 = 3$ unused codes.

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Concept check

1. When are two states equivalent?
2. Why can row matching miss an equivalence?
3. How many flip-flops does a 9-state machine need in binary? One-hot?
4. Why do JK designs often need fewer gates than D designs?

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Key takeaways

Equivalent

same outputs, equivalent next

Reduction

row match; implication table

Flip-flops

$2^n \geq$ states

D design

$D = Q^+$

T design

$T = Q \oplus Q^+$

JK design

excitation table, X's

Diagram, table, maps, circuit, check: every time.

Homework #9 due at the start of Lecture 21

This afternoon's lab: Lab Hp3 — Calculator 3: magnitude comparator

Next: registers, shift registers and ripple counters

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